TK12683

Profile Summary

- **Information Technology** student possessing a strong foundation in **Artificial Intelligence, Machine Learning and Backend Development.**
- Proficient in **Python,** with hands-on experience, AI development, working APIs, training ML models and backend applications development using FastAPI.
- Experienced in implementing machine learning workflows including data preprocessing, model training, evaluation, and deployment.
- Skilled in working with Large Language Models (LLMs), prompt engineering, and building AI agents for automation, chatbots, retrieval-augmented generation (RAG), and decision-support systems.
- Strong understanding of **Software Development Life Cycle (SDLC)**, Object-Oriented Programming (OOP), Data Structures, and Unit Testing.
- Skilled in version control using Git and GitHub, with the ability to translate e-business requirements into scalable technical solutions.

Technical Skills

- Programming Languages: **Python, SQL**
- AI Backend Frameworks & APIs: **Django, FastAPI, Postman** (**API Testing)**
- Databases: **PostgreSQL**
- AI & Machine Learning: **Scikit-Learn, Pandas, NumPy, Machine Learning Algorithms, Langchain, Large Language Model (LLM), RAGs, ChatBot development**
- Development Tools: **Git, GitHub, VS Code**
- Methodologies: **SDLC, Object-Oriented Programming (OOPS), Data Structures, Unit Testing**

Major Projects

1. **HEART SENSE – Risk Assessment & Prediction Model**

Duration: 3rd June 2025 - 22nd July 2025
Tech Stack: Python, Machine Learning, Django

Description:
Developed a machine learning-based attack risk prediction system that analyzes multiple health parameters to estimate the probability of heart attack.

Responsibilities:
- Implemented ML algorithms in Python for model training and prediction.
- Performed data preprocessing, feature selection, and model validation to ensure prediction reliability.
- Created and tested synthetic datasets to validate model performance.
- Built and deployed the application using Django to provide a user-friendly web interface.
- Documented model performance metrics and technical implementation details.

2. **RESUME ANALYZER - Rag Application**
   TechStack:
   - **Backend:** FastAPI, Uvicorn, Pydantic
   - **AI/ML:** OpenAI Embeddings, Groq, Langchain
   - **Vector DB:** FAISS

Responsibilities:
- Designed and implemented a RAG-based pipeline to compare resumes with job descriptions by extracting, preprocessing, and chunking text data.
- Generated **semantic embeddings using OpenAI** and stored job description vectors in **FAISS** for efficient similarity search.
- Built a **retrieval system** to fetch relevant job requirements and used a **Groq-powered LLM** to generate match scores, skill gaps, and improvement suggestions.
- Developed a **FastAPI backend** to handle file uploads, run the pipeline, and return structured analysis results via REST APIs.

3. **DAM System (Digital Asset Management) - An AI powered platform**
   Techstack:
   - Backend: Python, FastAPI, SQLAlchemy, Uvicorn
   - **AI/ML:** Sentence Transformer, Groq, Langchain
   - **Asset/Metadata Storage:** Cloudinary, PostgreSQL
   - **Authentication:** JWT Authentication
   - **Version Control:** GitHub, Git

Description:
Developed a full-stack AI-powered Digital Asset Management (DAM) platform for centralized storage, governance, review, and intelligent management of enterprise digital assets. The platform supports role-based workflows, AI-assisted metadata generation, automated asset review pipelines, and cloud-based asset delivery.

Responsibilities:

- Designed and developed an AI-powered Digital Asset Management platform enabling centralized storage, organization, review, approval, and distribution of digital assets including images, videos, documents, and marketing content.
- Built secure role-based access control (RBAC) workflows supporting Super Admin, Admin, Reviewer, and User roles with permission-based asset management and approval processes.
- Developed scalable REST APIs using FastAPI for asset upload, metadata management, version control, approval workflows, user management, analytics, and notification services.
- Implemented cloud-based asset storage and delivery using Cloudinary, including automatic media optimization, thumbnail generation, preview handling, and secure asset retrieval.
- Designed asset lifecycle management workflows including draft creation, review queues, approval/rejection pipelines, version tracking, and audit-ready asset governance processes.
- Built intelligent metadata management features supporting asset categorization, ownership tracking, usage rights management, audience segmentation, campaign tagging, and business domain classification.
- Designed and implemented an AI-powered Semantic Retrieval and Hybrid Search engine enabling natural-language asset discovery through SentenceTransformers-based embeddings, ChromaDB vector search, metadata-aware semantic indexing, and intelligent asset enrichment. Built a hybrid retrieval architecture combining semantic similarity search with traditional metadata and keyword-based filtering, allowing users to discover assets using contextual queries, business concepts, OCR content, generated captions, tags, and descriptive natural-language prompts rather than exact filenames or metadata values.
- Developed notification and workflow automation services to inform reviewers and administrators about asset submissions, approvals, rejections, and system events.
- Designed relational database models and CRUD services using SQLAlchemy for users, assets, asset versions, notifications, review history, permissions, and activity tracking.
- Developed responsive React.js dashboards for asset management, review operations, analytics visualization, user administration, and role-specific workflows.
- Implemented modular backend architecture with reusable service layers, authentication middleware, database abstraction, and API-driven frontend integration to support future scalability and cloud migration.

4. AI-Powered Proposal Generator

◦ **Description:** Developed an AI-powered proposal generation platform that automatically creates structured business and technical proposals based on user requirements using AI-driven content generation.

◦ **Tech Stack:** FastAPI, Python, OpenAI API, GitHub

◦ **Responsibilities:**

▪ Designed and implemented AI-powered proposal generation workflows for creating business and technical proposals.

▪ Developed prompt engineering pipelines to generate proposal sections such as objectives, deliverables, technical approach, technology stack, and future scope.

▪ Built FastAPI backend services to manage proposal generation requests and orchestrate AI workflows.

▪ Integrated OpenAI APIs for dynamic content generation and context-aware proposal creation.

▪ Implemented proposal export functionality for generating structured proposal documents and PDFs.

▪ Managed version control and collaborative development using GitHub.

Internship Experience

- **Machine Learning Intern**
  - Trained in Machine Learning using Python; implemented algorithms for data pre-processing, model training, and performance evaluation.
  - Created Technical Reports, summarizing model accuracy and data insights for project reviews.

EDUCATION

- B.Tech (Information Technology) from MCKV Institute Of Engineering, Howrah.